

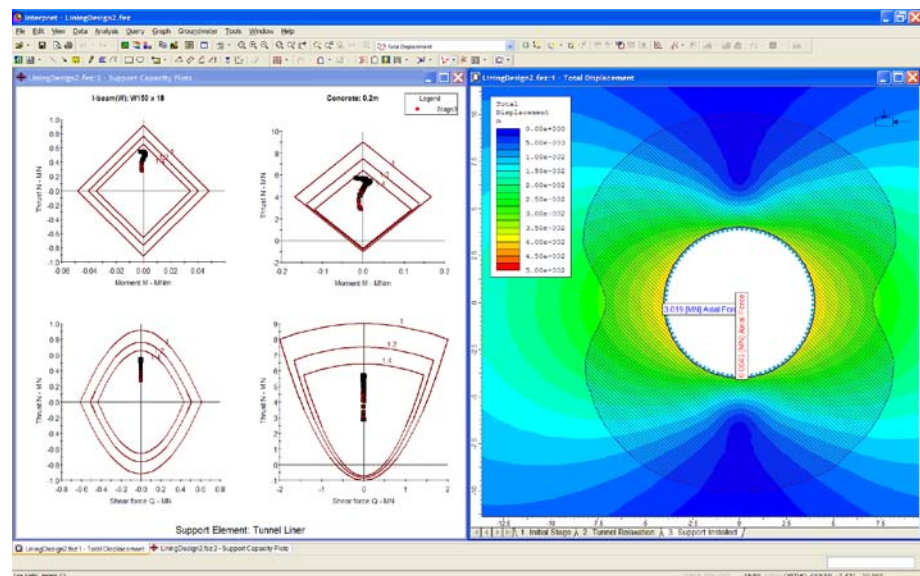
Tunnel Lining Design

In this tutorial, *Phase2* is used to design a reinforced concrete tunnel liner.

The complete model can be found in the **Tutorial 24 Tunnel Lining Design.fez** file. All tutorial files installed with *Phase2* 8.0 can be accessed by selecting File > Recent Folders > Tutorials Folder from the *Phase2* main menu.

Topics covered

- Reinforced concrete liners
- 3D tunnel simulation
- Distributed loads – field stress vector option
- Support capacity curves
- GSI calculator



Problem

A circular tunnel of radius 4m is to be constructed in Schist at a depth of 550m. The in-situ stress field has been measured with the major in-plane principal stress equal to 30 MPa, the minor in-plane principal stress equal to 15 MPa and the out-of-plane stress equal to 25 MPa. The major principal stress is horizontal and the minor principal stress is vertical. The strength of the Schist can be represented by the Generalized Hoek-Brown failure criterion with the uniaxial compressive strength of the intact rock equal to 50 MPa, the GSI equal to 50 and m_i equal to 10. To compute the rock mass deformation modulus, the modulus ratio (MR) is assumed to be 400. The support is to be installed 2m from the tunnel face.

The goal of this tutorial is to design a reinforced concrete lining with a factor of safety greater than 1.4.

To design a support system, the following three steps must be performed:

1. Determine the amount of tunnel wall deformation prior to support installation. As a tunnel is excavated, there is a certain amount of deformation, usually 35-45% of the final tunnel wall deformation, before the support can be installed. Determining this deformation can be done using either a) observed field values, or b) numerically from 3D finite-element models or axisymmetric finite-element models, or c) by using empirical relationships such as those proposed by Panet or Vlachopoulos and Diederichs.
2. Using either the internal pressure reduction method, or the modulus reduction method (see tutorial 18), determine the internal pressure or modulus that yields the amount of tunnel wall deformation at the point of and prior to support installation. This is the value determined in step 1.
3. Build a model that relaxes the boundary to the calculated amount in step 2 using either an internal pressure or modulus. Add the support and determine whether a) the tunnel is stable, b) the tunnel wall deformation meets the specified requirements, and c) the tunnel lining meets certain factor of safety requirements. If any of these conditions are not met, choose a different support system and run the analysis again.

Model

The first step is to determine the amount of tunnel wall deformation prior to support installation. For this tutorial, we'll use the relationship proposed by Vlachopoulos and Diederichs. The Vlachopoulos and Diederichs method is documented in Appendix 1 of the Kersten Lecture by Hoek, Carranza-Torres, Diederichs and Corkum. The paper is in the Hoek's published papers area on the Rocscience website:

<http://www.roscience.com/hoek/references/Published-Papers.htm>

This method requires that we build a model of the tunnel and determine a) the deformation far from the tunnel face using a simple plane strain analysis, and b) for the same model determine the plastic zone radius.

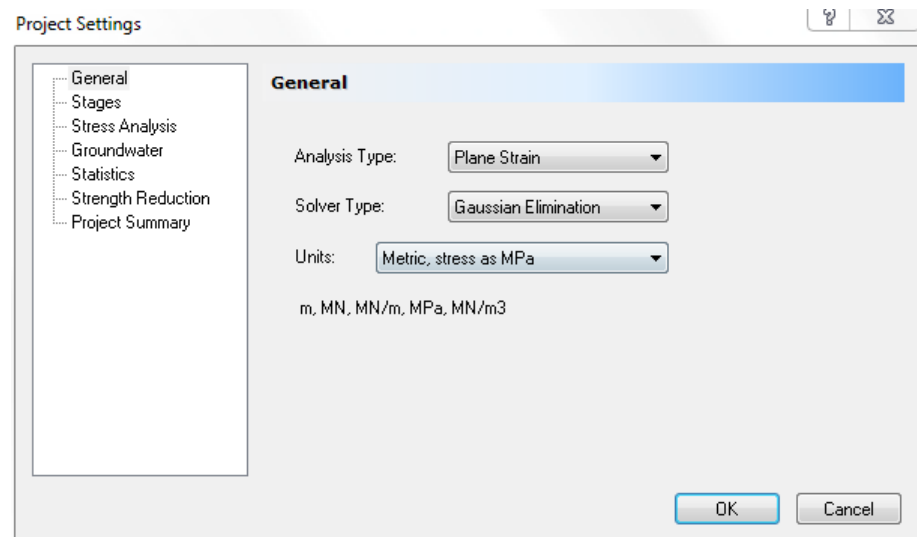
In this tutorial we'll start by building a single model that also combines step 2 with step 1. We'll build a plane strain model that relaxes an internal pressure on the tunnel boundary from a value equal to the applied in-situ stress to zero. The final stage, with zero internal pressure, will be used to determine the amount of deformation prior to support installation (step 1). The factoring of the applied internal pressure over a number of stages will be used to determine the pressure that yields the amount of tunnel wall deformation at the point of support installation (step 2).

Start the *Phase2* Model program.

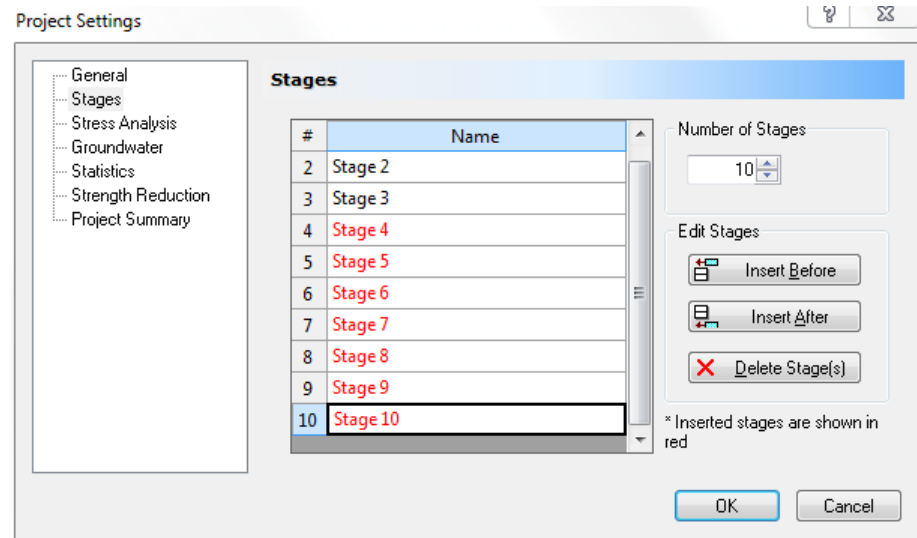
Project Settings



Open the **Project Settings** dialog from the **Analysis** menu and make sure the General tab is selected. Define the units as being “Metric, stress as MPa”.



Select the Stages tab. Change the number of stages to 10 (see following figure). Close the dialog by clicking OK.



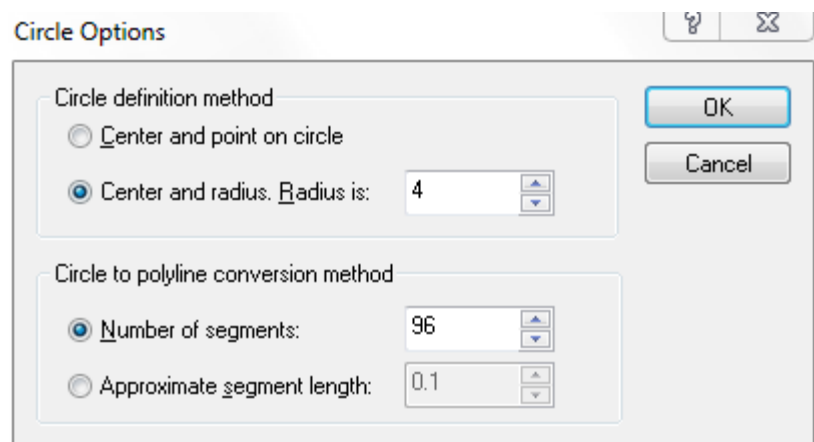
Geometry

Now enter the circular tunnel.



Select: Boundaries → Add Excavation

1. Right-click the mouse and select the Circle option from the popup menu. You will see the following dialog.



2. Select the Center and radius option, enter Radius = 4 and enter Number of Segments = 96 and select OK.
3. You will be prompted to enter the circle center. Enter 0,0 in the prompt line, and the circular excavation will be created.



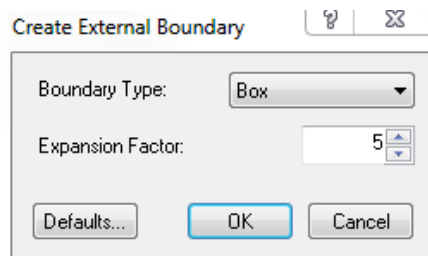
Select Zoom All (or press the F2 function key) to zoom the excavation to the center of the view.

Now we will create the external boundary. In *Phase2*, the external boundary may be automatically generated, or user-defined. We will use one of the 'automatic' options.



Select: Boundaries → Add External

You will see the Create External Boundary dialog. We will use the settings of Boundary Type = Box and Expansion Factor = 5. Select OK, and the external boundary will be automatically created.

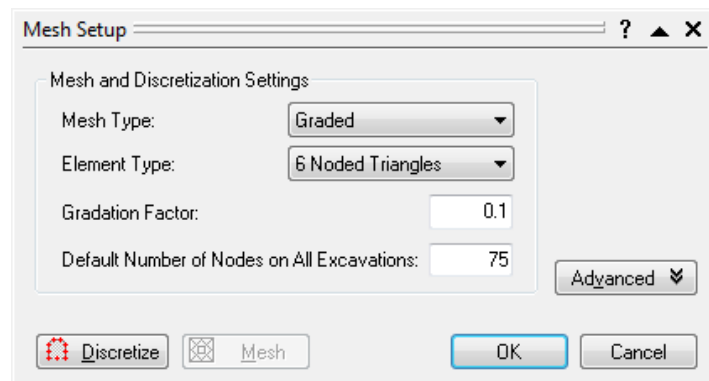


The boundaries for this model have now been entered.

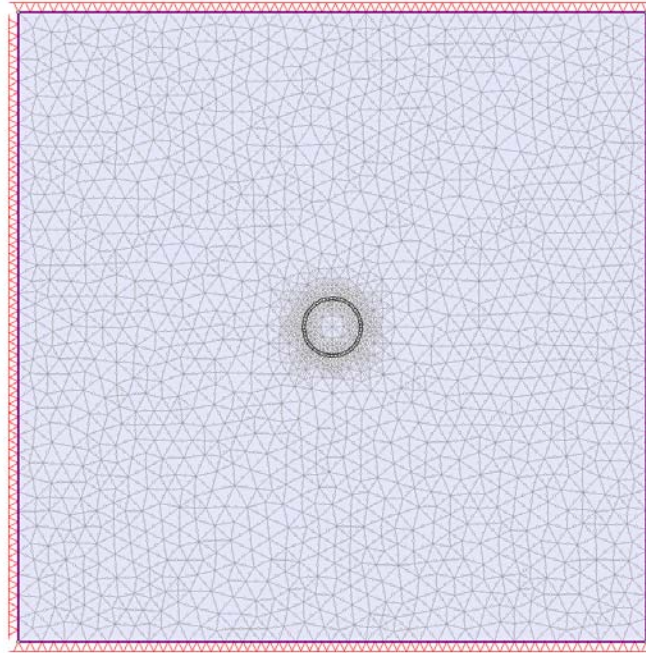
Mesh



Add the finite element mesh by selecting **Mesh Setup** from the **Mesh** menu. In the mesh setup dialog, change the Element Type to 6 Noded Triangles.



Click the Discretize button and then the Mesh button. Click OK to close the dialog. The mesh will look like this:



Boundary Conditions

For this tutorial, no boundary conditions need to be specified by the user. The default boundary condition will therefore be in effect, which is a fixed (i.e. zero displacement) condition for the external boundary.

Field Stress

Field Stress determines the initial in-situ stress conditions, prior to excavation. As described earlier in this tutorial, the in-situ stress field has been measured with the major in-plane principal stress equal to 30 MPa, the minor in-plane principal stress equal to 15 MPa and the out-of-plane stress equal to 25 MPa. The major principal stress is horizontal and the minor principal stress is vertical.

Select: Loading → Field Stress

Field Stress Properties

Field Stress Type: **Constant**

Sigma 1 (MPa, Comp. +): 30

Sigma 3 (MPa, Comp. +): 15

Sigma Z (MPa, Comp. +): 25

Angle (degrees from horizontal, CCW): 0

Locked-in horizontal stress (in plane) (MPa, Comp. +): 0

Locked-in horizontal stress (out-of-plane) (MPa, Comp. +): 0

OK Cancel Statistics... Advanced >>

Enter Sigma 1 = 30, Sigma 3 = 15, Sigma Z = 25, Angle = 0, and select OK.

Materials

Go to the **Properties** menu and select **Define Materials**.



For Material 1, change the Failure Criterion to Generalized Hoek-Brown and the Material Type to Plastic. Now define the strength parameters and the Young's Modulus using the GSI calculator. Press the GSI calculator button (see below).

Define Material Properties

Material 1

Name: Material 1 Material Color: [Color Selection]

Initial Element Loading: **Field Stress Only** Unit Weight: (MN/m3): 0.027

Elastic Properties

Elastic Type: **Isotropic** Poisson's Ratio: 0.3

Young's Modulus (MPa): 6143.7 ☒ Young's Modulus (resid) (MPa): 20000

E1 (MPa): 20000 E2 (MPa): 20000 Ez (MPa): 20000

v1z: 0.2 v1z: 0.2 v2z: 0.2

Strength Parameters

Failure Criterion: **Generalized Hoek-Brown**  Material Type: **Plastic**

Intact Comp. Strength (MPa): 50 Dilatation Parameter: 0

mb Parameter (peak): 1.67677 mb Parameter (resid): 1.67677

s Parameter (peak): 0.003866 s Parameter (resid): 0.003866

a Parameter (peak): 0.505734 a Parameter (resid): 0.505734

☐ Stage Properties ☐ Datum Dependent

Define Factors... Define Properties...

Unsaturated Shear Strength

Phi b: 0 Air Entry (MPa): 0

Copy To... Statistics... ☒ Show only properties used in model OK Cancel

In the GSI calculator dialog, set the uniaxial compressive strength of the intact rock equal to 50 MPa, the GSI equal to 50 and m_i equal to 10. To compute the rock mass deformation modulus, set the modulus ratio (MR) to 400. The dialog should look like:

Parameter Calculator

Intact UCS (MPa): 50

Geological Strength Index: 50

Intact Rock Constant m_i : 10

Disturbance Factor: 0

m_b : 1.677

s : 0.0038659

a : 0.5057

☒ Compute rock mass elastic modulus

Method: Generalized Hoek-Diederichs

☐ E_i (MPa): 20000.0

☒ MR: 400

Erm (MPa): 6143.7

☒ Apply m_b, s, a to peak properties

☒ Apply m_b, s, a to residual properties

OK Cancel

Press the OK button. The material properties dialog should now be updated with the new strength and modulus values.

Define Material Properties

Material 1

Name: Material 1 Material Color: [Color Picker]

Initial Element Loading: Field Stress Only Unit Weight: (MN/m3): 0.027

Elastic Properties

Elastic Type: Isotropic Poisson's Ratio: 0.3

Young's Modulus (MPa): 6143.7 ☐ Young's Modulus (resid) (MPa): 20000

E_1 (MPa): 20000 E_2 (MPa): 20000 E_3 (MPa): 20000

ν_{12} : 0.2 ν_{13} : 0.2 ν_{23} : 0.2

Strength Parameters

Failure Criterion: Generalized Hoek-Brown Material Type: Plastic

Intact Comp. Strength (MPa): 50 Dilation Parameter: 0

m_b Parameter (peak): 1.67677 m_b Parameter (resid): 1.67677

s Parameter (peak): 0.003866 s Parameter (resid): 0.003866

a Parameter (peak): 0.505734 a Parameter (resid): 0.505734

☐ Stage Properties ☐ Datum Dependent

Define Factors... Define Properties...

Unsaturated Shear Strength

Φ b: 0 Air Entry (MPa): 0

Copy To... Statistics... ☐ Show only properties used in model

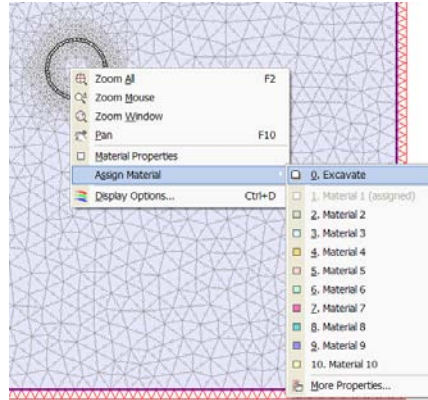
OK Cancel

Click OK to close the dialog.

Since you entered properties with the first (Material 1) tab selected, you do not have to Assign these properties to the model. *Phase2* automatically assigns the Material 1 properties for you.

Excavation

The tunnel is to be excavated in the first stage so click on the Stage 1 tab at the bottom of the screen. Simply place the mouse pointer inside the excavation and right-click the mouse. From the menu that pops up, select the **Assign Material > Excavate** option.



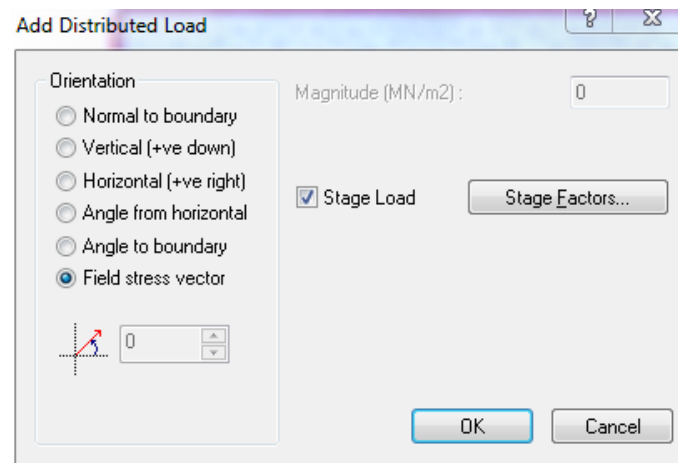
The material inside the excavation should now be removed.

Adding an Internal Pressure to the Excavation

Now let's add a uniform distributed load to the tunnel in stage 1. The magnitude and direction of the load will be equal and opposite to the in-situ stresses thus forming a balance between the stresses in the rock and the pressure inside the tunnel. Since the pressure is equal and opposite to the in-situ stress, no deformation should occur. However, in stage 2 and after, we will factor the load and gradually reduce the magnitude of the pressure. As a result, tunnel deformation will increase as the pressure is lowered to zero.

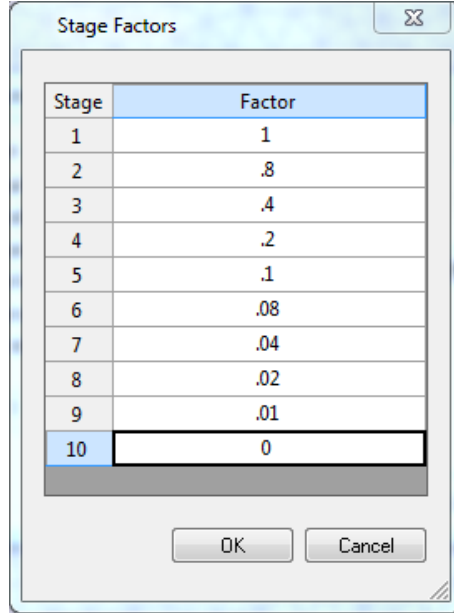


Select: Loading → Distributed Loads → Add Uniform Load



In the Add Distributed Load dialog, select the **Field stress vector** orientation option. Select the Stage Load checkbox, and select the Stage Factors button.

In the Stage Factors dialog enter the factors shown in the following image.



Stage	Factor
1	1
2	.8
3	.4
4	.2
5	.1
6	.08
7	.04
8	.02
9	.01
10	0

Factor = 1 means the magnitude will be the same as the field stress while a Factor = 0 means no load will be applied at that stage. Other values of Factor can be used to increase or decrease the magnitude of a load at any stage of a model.

Select OK in both dialogs. You will now be asked to pick the boundary segments on which the load will be applied.

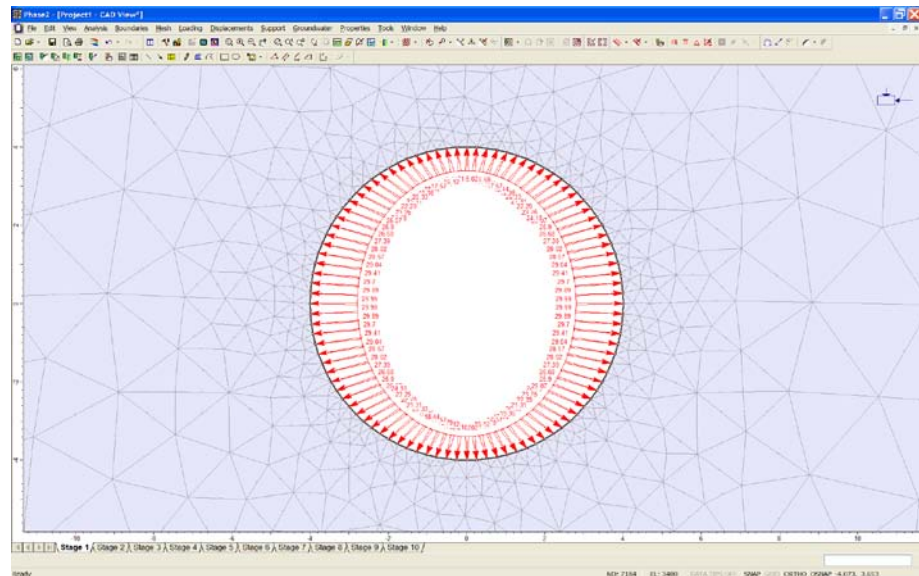
Select the excavation line segments to be loaded:

Select boundary segments [enter=done,esc=cancel] : **use the mouse to draw a selection window around the entire excavation. After the excavation segments are selected, right-click and select Done Selection, or press Enter.**

Note: to draw a selection window, simply pick one of the window corners by moving the mouse cursor to a point, and press AND HOLD DOWN the left mouse button. Now move the cursor while still holding down the left mouse button, you should see a window forming. Now move the mouse cursor to the opposite corner of the window and release the left mouse button when done.



Click the Zoom Excavation button on the toolbar. You should see the following:



Now click through the stage tabs. You should see the internal pressure reduce as the stage increases.

NOTE: a useful feature of the **Field stress vector** loading option is that any change to the field stress through the field stress dialog will also automatically update the internal pressure inside the tunnel.

We are now ready to run the analysis.

Compute

Before you analyze your model, let's save this as a new file called **InternalPressureReduction.fez**



Select: File → Save

Save the file as InternalPressureReduction.fez.



Select: Analysis → Compute

The *Phase2* Compute engine will proceed in running the analysis. When completed, you will be ready to view the results in Interpret.

Interpret

From Model, switch to the Interpret program.



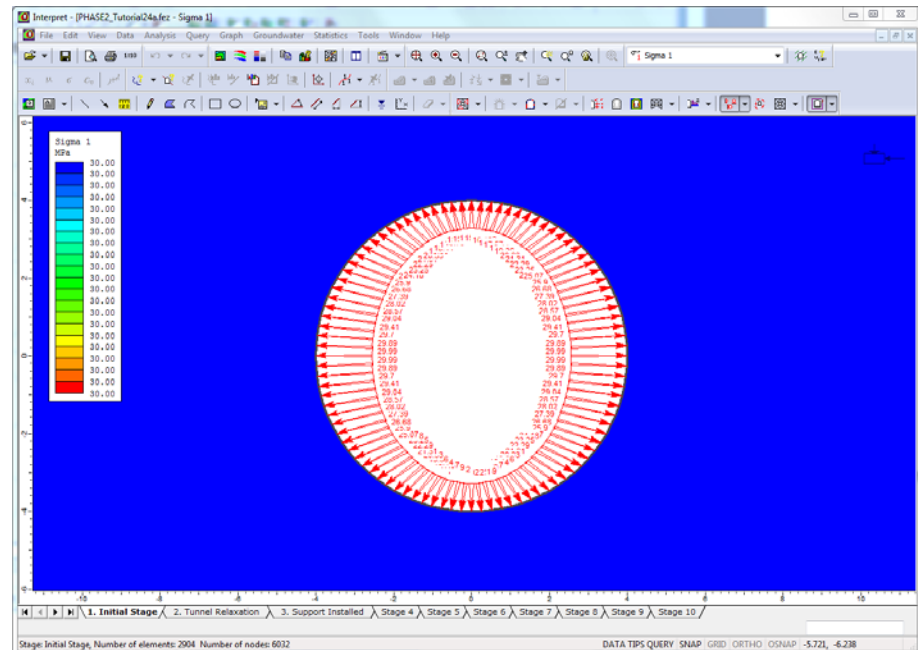
Select: Analysis → Interpret

After you select the Interpret option, the Interpret program starts and reads the results of the analysis. You will see the maximum stress, sigma 1 for Stage 1. Notice that there is no variation of stress and that the stress (30 MPa) is equal to the major in-situ field stress. This means that the internal pressure is equal and opposite to the field stress and the model is behaving as if the tunnel did not exist.



Now click the Zoom Excavation button on the toolbar.

Change the contours to plot Total Displacement using the pull down menu in the toolbar. The model for Stage 1 will look like this:



You can see that there no displacement in the first stage.

Now click through the stages. You'll see an increase in deformation around the tunnel as the internal pressure is reduced.

Step 1 – Computing tunnel deformation before support installation using the Vlachopoulos and Diederichs method

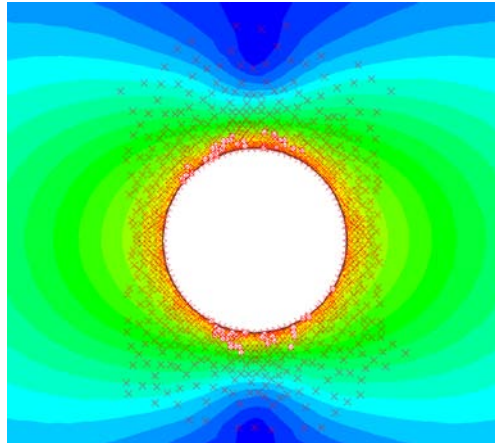
To compute the tunnel deformation at the point of support installation, we'll use the empirical relationship developed by Vlachopoulos and Diederichs. To use the Vlachopoulos and Diederichs method, you need two pieces of information from the finite-element analysis. You need to know a) the maximum tunnel wall displacement far from the tunnel face, and b) the radius of the plastic zone far from the tunnel face.

Both of these values can be computed from a plane strain analysis with zero internal pressure inside the excavation. In the model we just built, the results from stage 10 are used since there is zero internal pressure in this stage.

Switch to the last stage, stage 10. Look in the lower left corner of the program window on the status bar. You'll see that the maximum displacement for this stage is approximately 0.065m. This is the value of maximum wall displacement far from the tunnel face. The location of this displacement is in the roof and floor of the excavation. The location of this displacement is important since any comparisons of displacement for various internal pressures must be made at the same location.



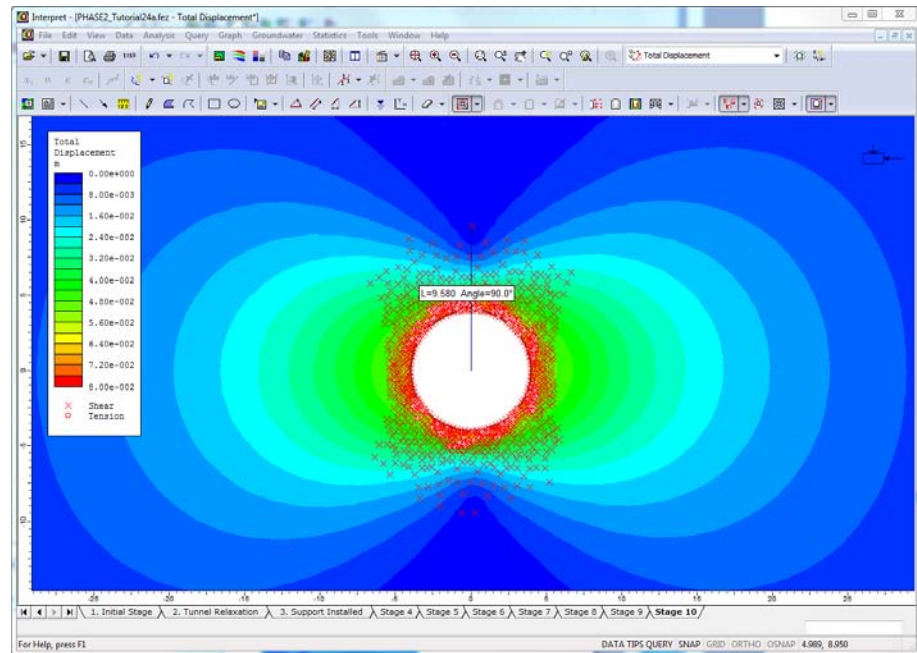
To determine the radius of the plastic zone, first turn on the display of yielded elements using the Display Yielded Elements  toolbar button. You'll see a number of crosses representing elements in the finite element analysis that have failed. **Zoom Out**  so that the entire extent of failed points is visible (see below).



The extent of this failed zone represents the extent of the plastic zone around the tunnel. To determine the radius of the plastic zone, you can use either the measuring tool or the dimensioning tool to measure the distance from the center of the tunnel to the perimeter of the yielded/plastic zone. In this tutorial we'll use the measuring tool.

Select: Tools → Add Tool → Measure

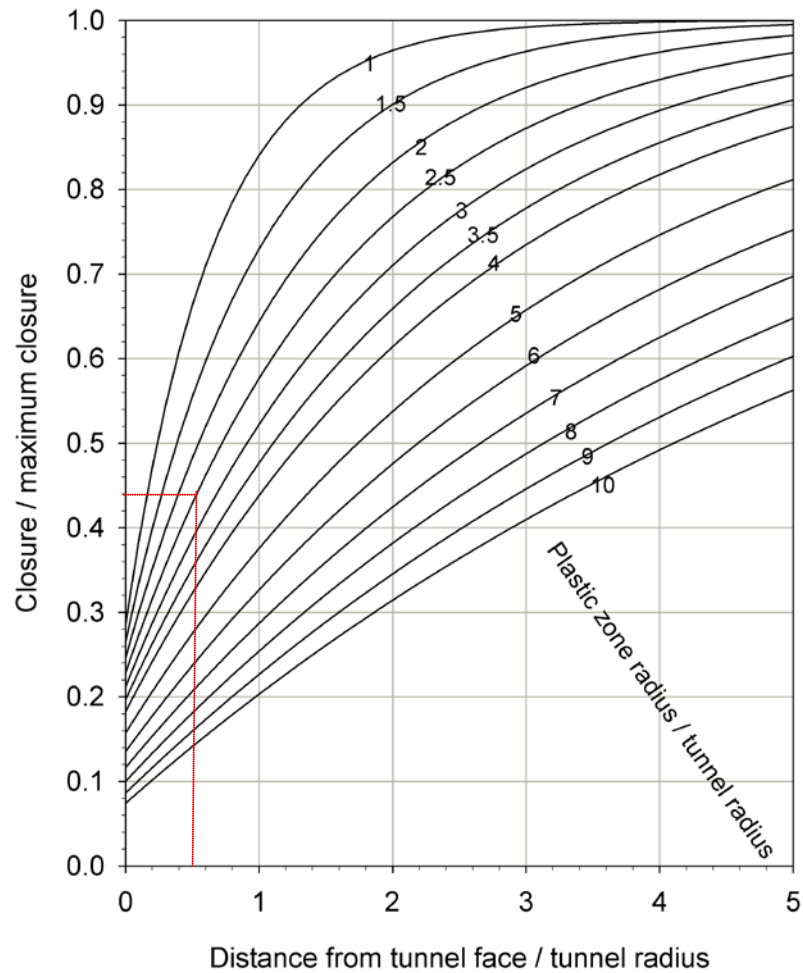
Pick the location to measure from [esc=quit]: 0,0
 Pick the location to measure to [esc=quit]: ***use the mouse to extend the measuring line vertically until you get to the edge of the yield zone, press the left mouse button.***



As seen above, the radius of the plastic zone is approximately 9.5m.

Computing displacement prior to support installation using the Vlachopoulos and Diederichs Method

The following plot was created using the Vlachopoulos and Diederichs equations. The equations can be found in the Kersten Lecture, appendix 1. Using this plot, you can easily estimate the amount of closure prior to support installation if you know the plastic radius and displacement far from the tunnel face.



For our problem, $R_p=9.5\text{m}$, $R_t=4\text{m}$, $X=2\text{m}$, and $u_{\max}=0.065\text{m}$. The Distance from tunnel face/tunnel radius = $2/4 = 0.5$. The Plastic zone radius/tunnel radius = $9.5/4 = 2.4$. From the above plot this gives Closure/max closure approximately equal to 0.44. Therefore the closure equals $(0.44) \cdot (0.065) = 0.028\text{m}$.

As computed above, the tunnel roof displaces 0.028m before the support is installed.

Step 2 - Determining the internal pressure factor

The next step is to determine the internal pressure that yields a displacement of 0.028m in the roof of the tunnel. It is important to maintain the same location as is used to determine u_{max} , since the location of maximum displacement can change depending on the magnitude of the internal pressure. This can be seen in this model as larger internal pressures produce larger displacement in the sidewall while smaller internal pressures produce larger displacements in the roof and floor.

To determine the internal pressure that yields a 0.028m roof displacement, we'll plot the displacement versus stage for a point on the roof of the excavation.

Make sure you have Total Displacement selected as the data type.

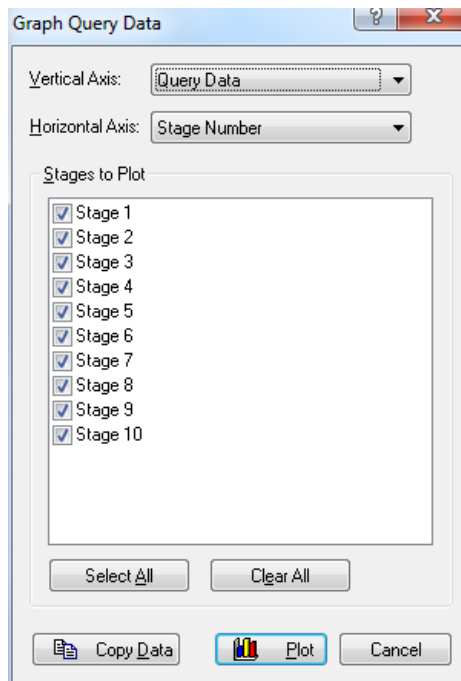
Graphing Displacement in the Roof of the Excavation

To create the graph:

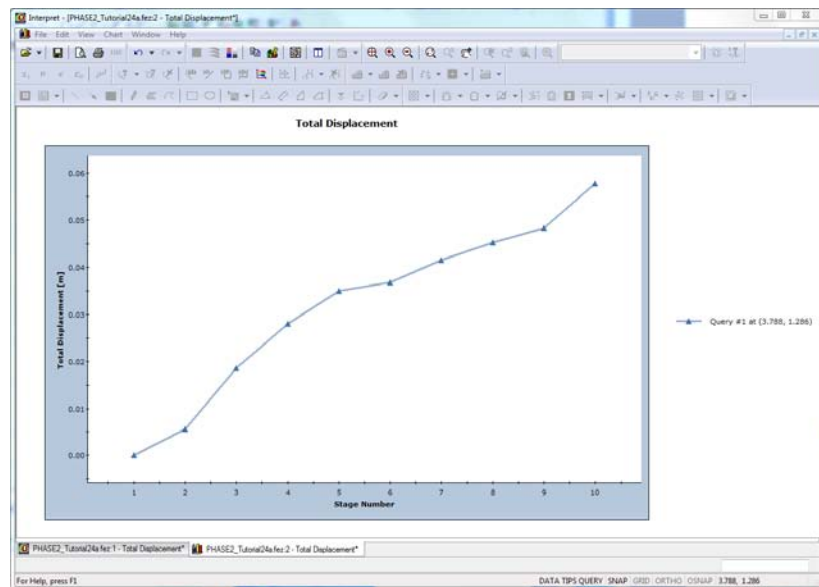


Select: Graph → Graph Single Point vs. Stage

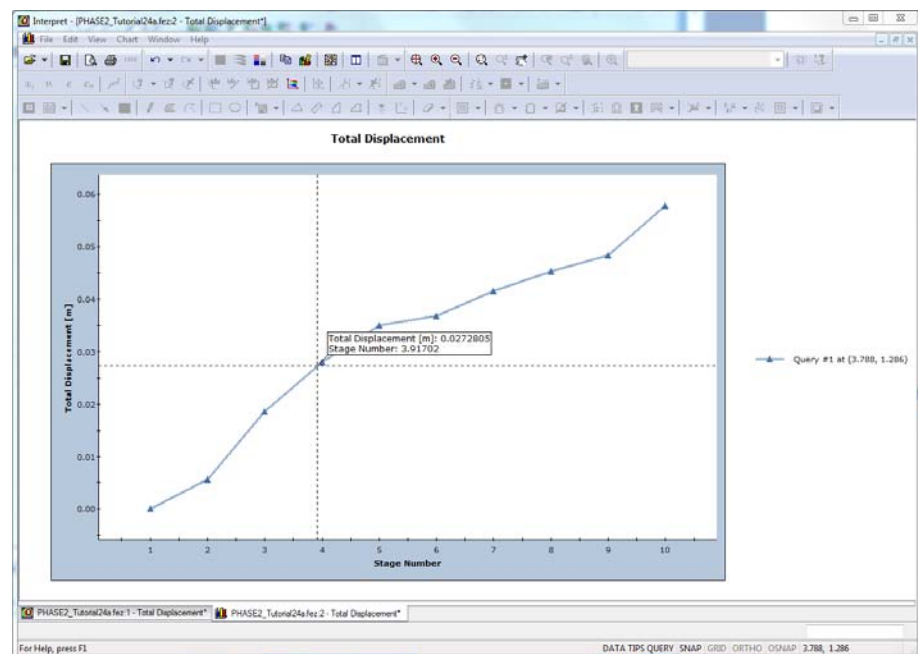
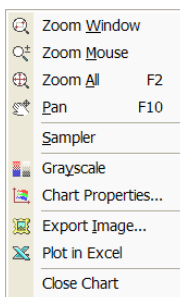
1. When asked to enter a vertex, type in the value 0,4 for the location and press Enter. This is a point on the roof of the excavation.
2. You will see the Graph Query Data dialog.



- Press the Create Plot button. The following figure shows the plot generated by the program. This is a plot of displacement versus stage for a point in the roof of the tunnel.



Right-click in the plot and choose the **Sampler** option. Move the sampler by moving the mouse with the left mouse button. Move the sampler until the displacement value on the right side of the plot is equal to 0.028m.



From this plot, you can see that in stage 5, the wall displacement in the roof of the tunnel is 0.028m. This represents an internal pressure factor of 0.1 as was defined in the modeler for the field stress vector distributed load.

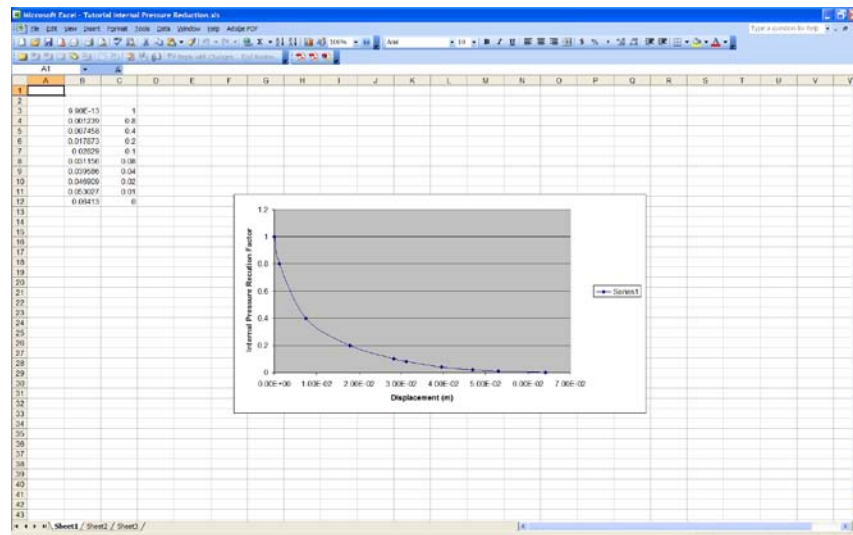
Creating a convergence confinement graph in Excel

Often you want to create a convergence confinement graph which plots displacement versus internal pressure. This is easily done by exporting the above graph to Microsoft Excel™. This requires that you have Excel installed on your computer.

Right-click in the Graph you just created and choose the **Plot in Excel** option.

Excel will launch with a plot of stage number versus displacement. You can easily modify the plot to change the stage number data to the internal pressure factor. A sample of the Excel file for this example is included in the Tutorials folder with the *Phase2* data files.

The following image shows the convergence-confinement plot in Excel for this example. You can see by this plot that an internal pressure factor of 0.1 yields the tunnel wall displacement computed above for the point of support installation (0.028m).



We have now completed steps 1 and 2 as defined in the Problem section at the beginning of this tutorial. It is now time to actually design our support system.



From Interpret, switch back to the *Phase2* Model program by pressing the **Model** button on the toolbar.

Model

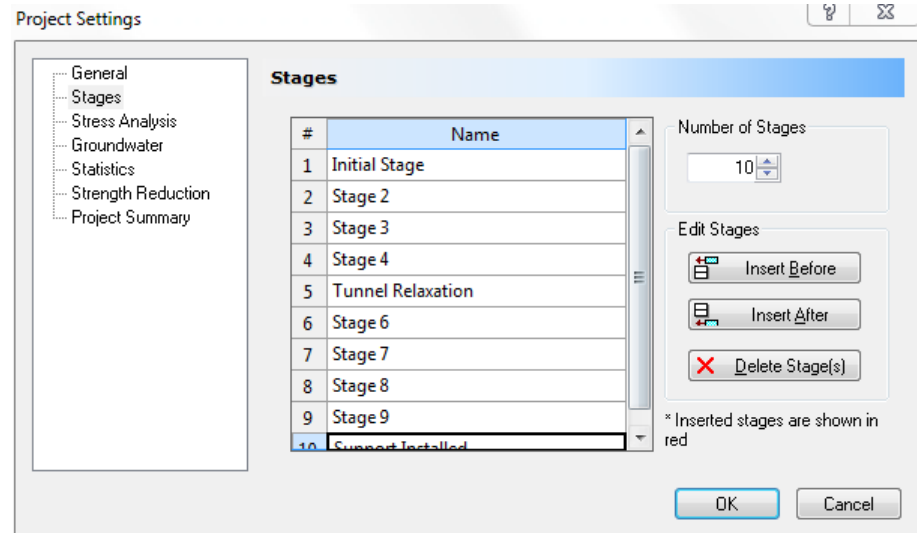
You should now be in the *Phase2* Model program with the 10 stage model you created above loaded into the program.

We will use this file and modify it to do the support design.

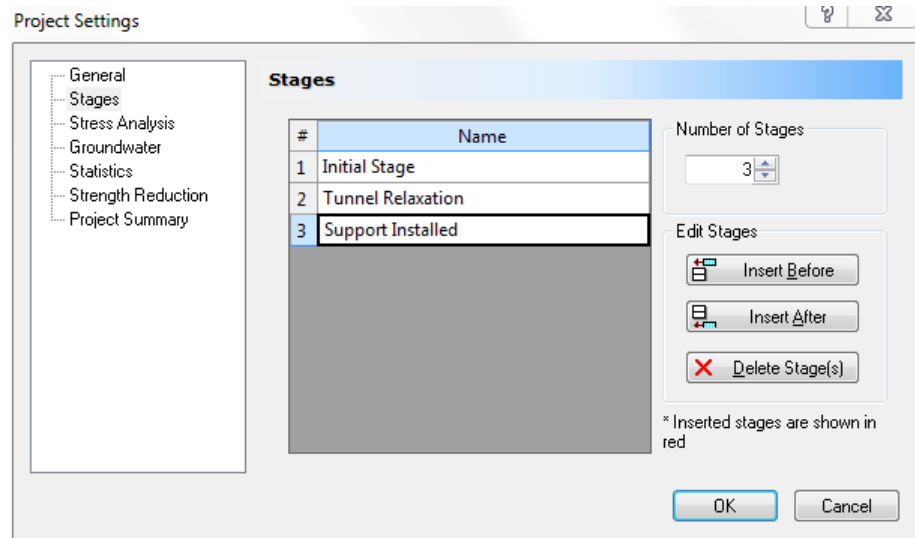
Project Settings



Open the **Project Settings** dialog from the **Analysis** menu and select the Stages Tab. Change the name of stage 1 to *Initial Stage*. Change the name of Stage 5 to *Tunnel Relaxation*. Change the name of Stage 10 to *Support Installed*. The dialog should look like this:



Now delete all other stages except these three stages (i.e. stages 2,3,4,6,7,8,9). Note, you can select multiple stages by scrolling down the number column with the left mouse button depressed. Use the Delete Stages button to delete the stages. After deleting these stages, the dialog should look like:

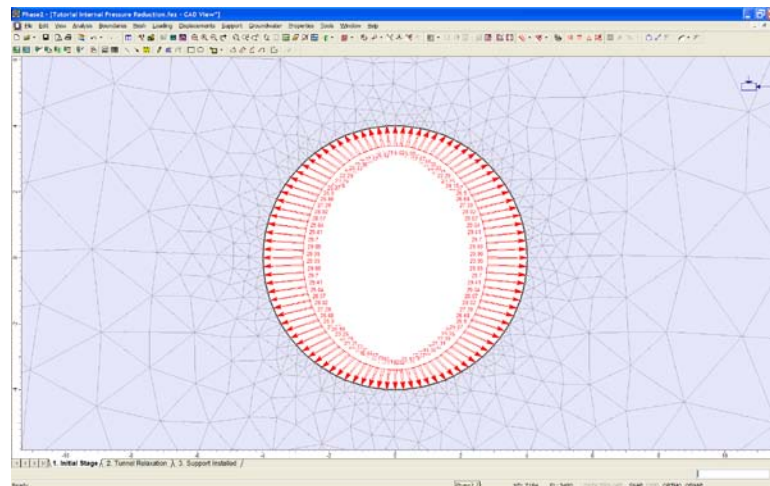


We chose stage 5 from the old model because it represents the stage in which the internal pressure in the tunnel yields the necessary deformation before we install the support. Close the dialog by clicking OK.

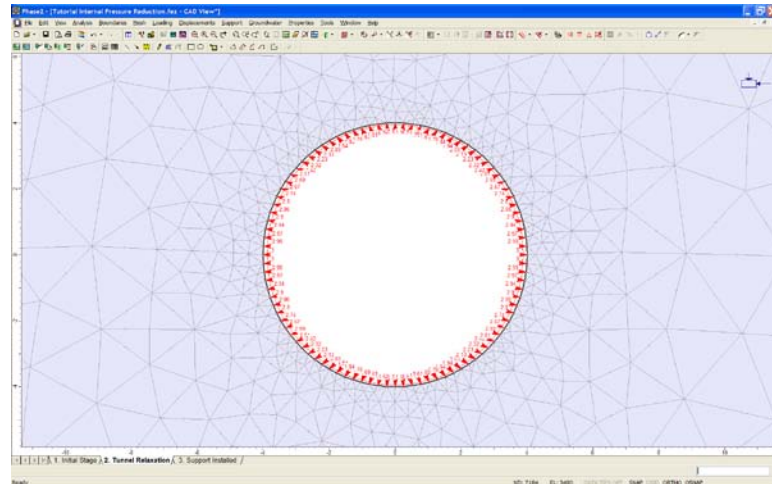


Make sure the Stage 1 tab is selected. Click the Zoom Excavation button on the toolbar.

You should see the following:



Click through the stages. Stage 2, the tunnel relaxation stage, should look like:



Note: you can use the Loading→Distributed Loads→Edit Distributed Load option to select any of the loads on the boundary to verify that the stage factor is 0.1 for Stage 2.

Stage 3, the Support Installed stage should have no load on the boundary.

Setting the Reinforced Concrete Liner Properties

Now define the liner properties. The properties we enter will correspond to a 100 mm thick layer of concrete reinforced with W100X19.3 I-beams spaced at 2 meter intervals along the tunnel axis.



Select: Properties → Define Liners

1. Change the Name of the liner to Tunnel Liner
2. Change the Liner Type to **Reinforced Concrete**
3. Click on the **Common Types** button. You will see the Reinforcement database dialog shown below. For the Reinforcement, we will select an I-beam from a list of standard reinforcement types.
4. In the Reinforcement database dialog, select the W100 x 19.3 I-beam. Click OK, and the I-beam reinforcement properties will be automatically loaded into the Define Liner Properties dialog.

Reinforcement

Shape: I-beam
 Lattice girder
 Hollow section
 Rebar
 Wire Mesh
 Channel
 Dbl Channel

Type: W
 M
 S
 HP
 HE Europe
 CP

Designation (Metric):
 W200 x 26.6
 W200 x 22.5
 W200 x 19.3
 W200 x 15
 W150 x 37.1
 W150 x 29.8
 W150 x 22.5
 W150 x 24
 W150 x 18
 W150 x 13.5
 W150 x 13
 W130 x 28.1
 W130 x 23.8
 W100 x 19.3

Section Depth (mm): 106
 Area (mm²): 2470
 Moment of Inertia (10e6mm⁴): 4.7
 Weight (kg/m): 19.3

☐ Imperial ☒ Metric

Edit Database... OK Cancel

5. In the Define Liner Properties dialog, for the Reinforcement, enter a spacing of 2m.
6. Enter the properties for the concrete. Thickness=0.1m, Modulus=25000MPa, Poisson Ratio=0.15, Compressive Strength=45MPa, Tensile Strength=5MPa. The liner properties dialog should look like:

Define Liner Properties

Tunnel Liner

Name: Tunnel Liner Color: ■

Liner Type: Reinforced Concrete

☒ Reinforcement
 Spacing (m): 2
 Section Depth (m): 0.106
 Area (m²): 0.00247
 Moment of Inertia (m⁴): 4.7e-006
 Young's Modulus (MPa): 200000
 Poisson Ratio: 0.25
 Compressive Strength (MPa): 400
 Tensile Strength (MPa): 400
 Weight (kg/m): 19.3
 I-beam(W): W100 x 19.3

☐ Stage Concrete Properties

☒ Concrete
 Thickness (m): 0.1
 Young's Modulus (MPa): 25000
 Poisson Ratio: 0.15
 Compressive Strength (MPa): 45
 Tensile Strength (MPa): 5
 Unit Weight (MN/m³): 0.024

Material Type: ☒ Elastic ☐ Plastic

☐ Include Weight in Analysis

☐ Sliding Gap
 Strain at Locking: 5 %

Beam Element Formulation: Timoshenko

Copy To... ☐ Show only properties used in model OK Cancel

7. Press OK to save your input and exit the dialog.

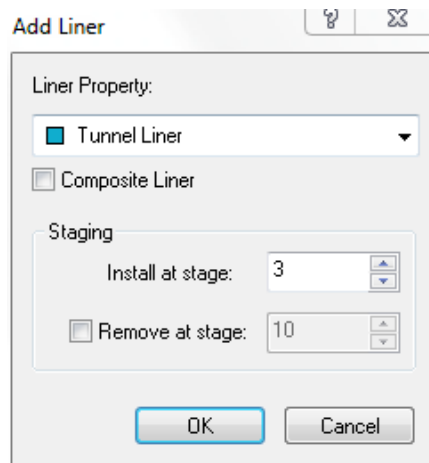
Adding a Reinforced Concrete Liner to the Tunnel

We will now line the tunnel with the liner defined above. First make sure that Stage 3, the Support Installed stage, is selected.

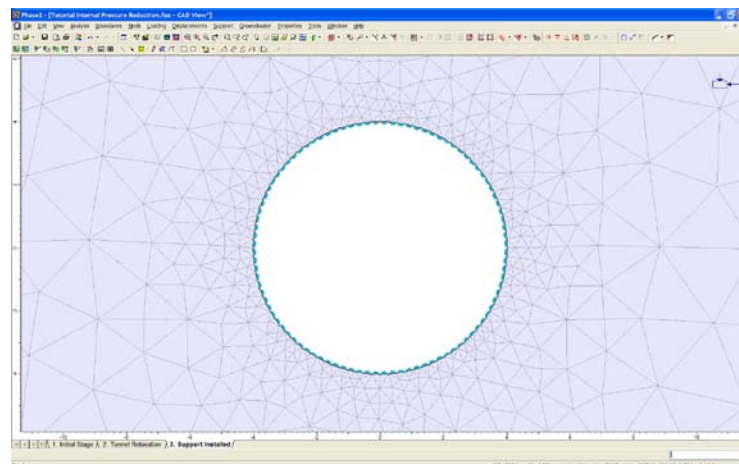


Select: Support → Add Liner

1. You will see the Add Liner dialog. Make sure it looks like the following image. Select OK.



2. Click and hold the left mouse button, and drag a selection window which encloses the entire excavation. Release the left mouse button. Notice that all excavation line segments are selected.
3. Right-click the mouse and select Done Selection, or just press the Enter key. The entire tunnel will now be lined, as indicated by the thick blue line segments around the excavation boundary (see below).



Click through the stages. Notice how the color of the liner changes from light blue in stages 1 and 2 to dark blue in stage 3. This indicates that the liner is being installed in stage 3.

We are now ready to run the analysis.

Compute

Before you analyze your model, let's save this as a new file called **LinerDesign.fez**. (Make sure you select Save As and not Save, or you will overwrite the internal pressure reduction file).

Select: File → Save As

Save the file as LinerDesign.fez.



Select: Analysis → Compute

The *Phase2* Compute engine will proceed in running the analysis. When completed, you will be ready to view the results in Interpret.

Interpret

From Model, switch to the Interpret program.



Select: Analysis → Interpret

If any other files are loaded into the Interpret program (i.e. the *InternalPressureReduction.fez* file), close them. Click on the tab at the bottom of the program window associated with the file and use the File→Close menu option to close the file.



Make sure the Stage 3 tab is selected. Click the Zoom Excavation button on the toolbar.

Support Capacity Diagrams

Support capacity diagrams give the engineer a method for determining the factor of safety of a reinforced concrete liner. For a given factor of safety, capacity envelopes are plotted in axial force versus moment space and axial force versus shear force space. Values of axial force, moment and shear force for the liner are then compared to the capacity envelopes. If the computed liner values fall inside an envelope, they have a factor of safety greater than the envelope value. So if all the computed liner values fall inside the design factor of safety capacity envelope, the factor of safety of the liner exceeds the design factor of safety.



Select: Graph → Support Capacity Plots

The Support Capacity Plot dialog allows you to choose the support element (i.e. liner type), the number of envelopes, and the stages from which the liner data is taken.

Use the spin control to increase the number of envelopes to 3. The dialog should look like:

Support Capacity Plot

Support Element:
Tunnel Liner

Factor of Safety Envelopes
Number of Envelopes: 3

#	Factor of Safety
1	1
2	1.2
3	1.4

Stages to Plot

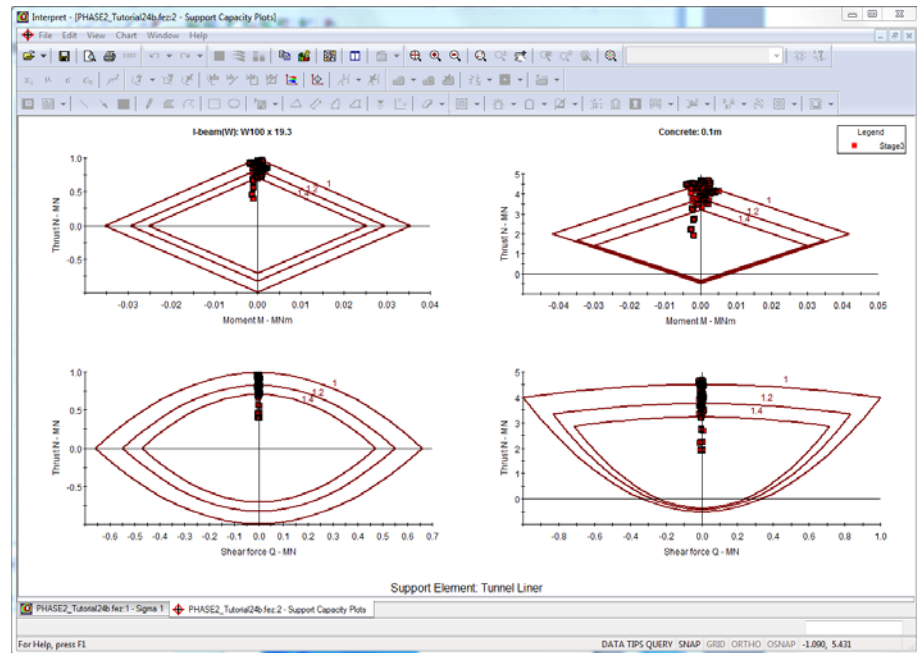
- ☐ Initial Stage
- ☒ Tunnel Relaxation
- ☒ Support Installed

Select All Clear All

OK Cancel

Press OK.

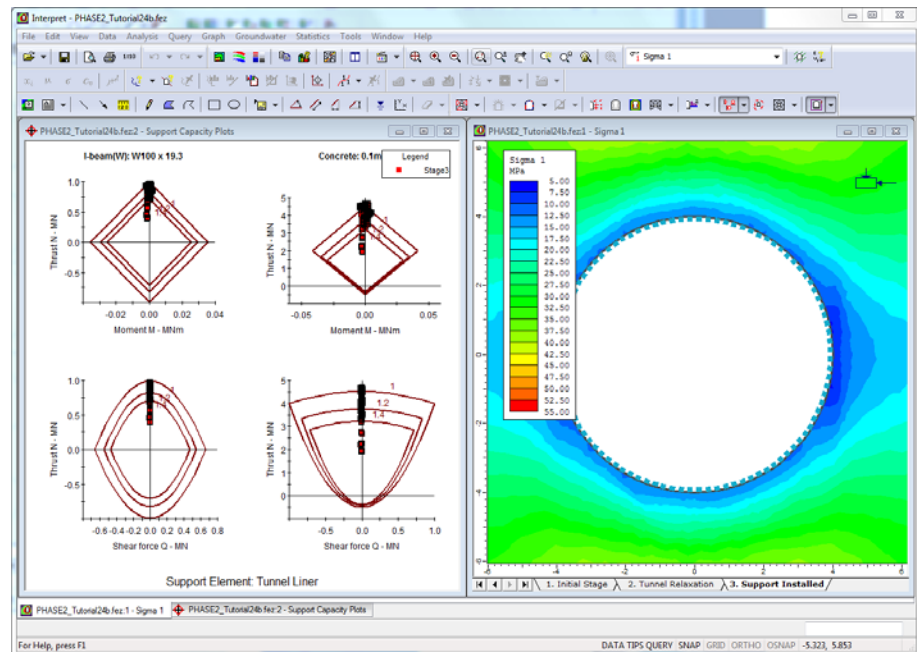
The following plot is generated. The dark red lines represent the capacity envelopes for the 3 factors of safety (1, 1.2, 1.4). Notice the number of liner data points that fall outside the 1.4 design factor of safety envelope, meaning they have a factor of safety less than 1.4. This occurs for both the capacity diagrams for the concrete and the capacity diagrams for the I-beam. In fact, a number of points fall outside the factor of safety=1.0 envelope. This liner would most likely experience cracking and crushing if used in this tunnel. Later, we'll have to improve on this design.



Let's investigate some of the things you can do with the support capacity envelopes.

Select: Window → Tile Vertically

The window should look like:

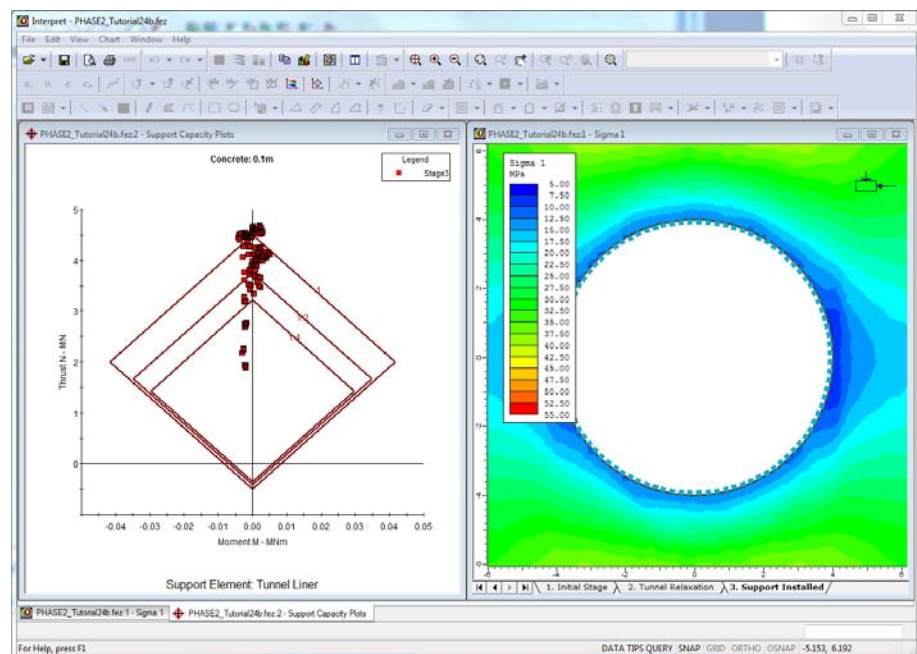


Make sure the Support Capacity Plot view is selected, not the contour view of the tunnel. If you are interested in just the concrete moment capacity plot, you can expand this plot using the following option.

Select: View → Concrete Moment Capacity Plot

The four plots are replaced by a single plot of the moment capacity for the concrete. Alternatively, you can maximize any single plot interactively by double-clicking on the plot. Double-clicking on the moment capacity for the concrete returns you to the four plots. Right-clicking also gives you a context menu that enables you to choose viewing options.

Make sure you have a single plot of the moment capacity for the concrete. Your display should look like:



Select: View → Zoom → Zoom Support Capacity Data

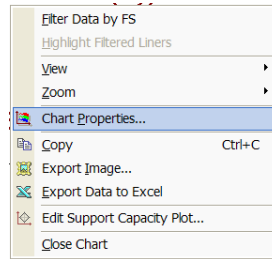
The view is zoomed so that the extents of the plot are determined by the extents of the moment and axial force data for the concrete.

Select: View → Zoom → Zoom All

The plot is returned to the default extents. If you have a mouse wheel, you can use it to zoom in and out on the data. Holding down the mouse wheel and moving the mouse results in panning of the plot. There are a number of options for manipulating the plot. Return to the default extents.

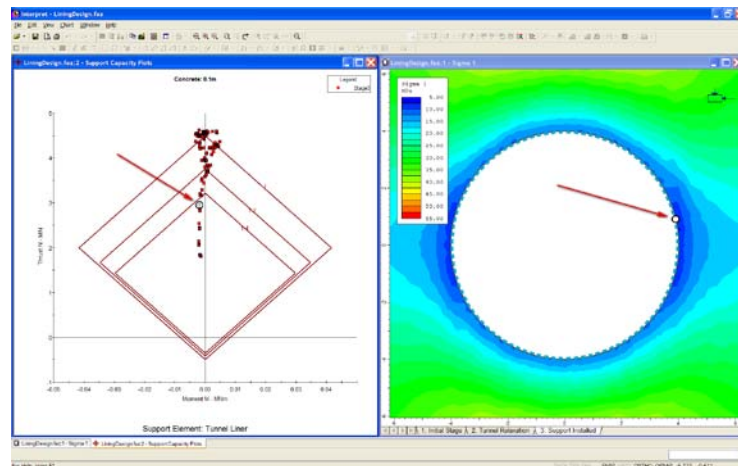
Select: View → Zoom → Zoom All

Try right-clicking in the plot view and choosing the Chart Properties option.

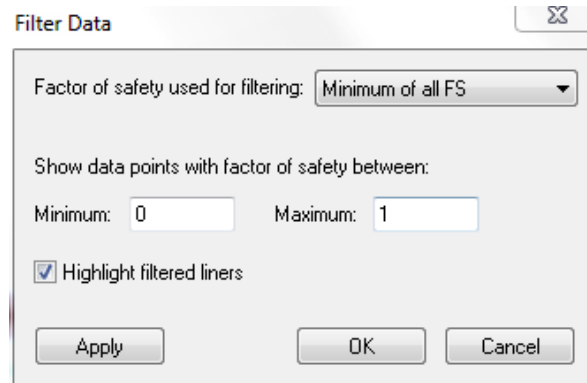


A dialog containing a number of options for changing the format of the plots is displayed. Close this dialog.

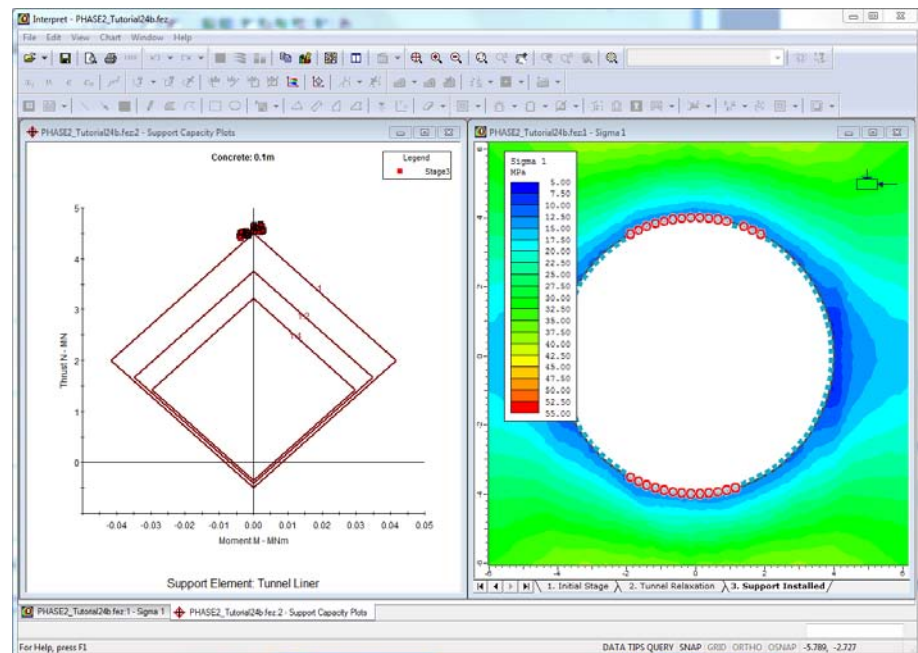
Now use the mouse to click on one of the red liner data points. The data point is highlighted in the support capacity plot view and the liner associated with this data is highlighted in the main contour view. This is shown in the following figure.



Right-click in the support capacity plot view and select the **Filter Data by FS** option. The following dialog is displayed. Change the Factor of safety used for filtering to **Concrete moment**. Change the Maximum value to 1 and turn on the Highlight filtered liners. What this does is plot all the data points with factor of safety between 0 and 1 for the concrete moment, and show the associated liner elements in the contour view.



Press the OK button after making these changes. In the following image, only the liner elements with factor of safety between 0 and 1 for the concrete are displayed. The liner elements associated with these data points are highlighted on the contour view by drawing a grey circle around each element. As you can see, the areas of minimum factor of safety for the concrete are in the roof and floor of the excavation.



Now let's go back to the Modeler and pick a better support system for the tunnel.

From Interpret, switch back to the *Phase2* Model program:



Select: Analysis → Model

Model – Improving the Support System



Select: Properties → Define Liners

1. Make sure the Tunnel Liner tab is selected. Click on the Common Types button.
2. In the Reinforcement dialog that is displayed, select the W150 x 18 I-beam. Click OK.
3. Increase the thickness of the concrete to 0.2m. The liner properties dialog should look like:

Define Liner Properties

Tunnel Liner

Name: Tunnel Liner Color: [Blue]

Liner Type: Reinforced Concrete

☒ Reinforcement **Common Types**

Spacing (m): 2

Section Depth (m): 0.153

Area (m²): 0.00229

Moment of Inertia (m⁴): 9.2e-006

Young's Modulus (MPa): 200000

Poisson Ratio: 0.25

Compressive Strength (MPa): 400

Tensile Strength (MPa): 400

Weight (kg/m): 18

I-beam(W): W150 x 18

☐ Stage Concrete Properties

☐ Concrete

Thickness (m): 0.2

Young's Modulus (MPa): 25000

Poisson Ratio: 0.15

Compressive Strength (MPa): 45

Tensile Strength (MPa): 5

Unit Weight (MN/m³): 0.024

Material Type: ☒ Elastic ☐ Plastic

☐ Include Weight in Analysis

☐ Sliding Gap

Strain at Locking: 5 %

Beam Element Formulation: Timoshenko

Copy To... ☐ Show only properties used in model

OK Cancel

4. Press OK to exit the dialog.

We are now ready to re-run the analysis.

Compute

Before you analyze your model, let's save this as a new file called **LinerDesign2.fez**. (Make sure you select Save As and not Save, or you will overwrite the **LinerDesign.fez** file).

Select: File → Save As

Save the file as LinerDesign2.fez.



Select: Analysis → Compute

The *Phase2* Compute engine will proceed in running the analysis. When completed, you will be ready to view the results in Interpret.

Interpret

From Model, switch to the Interpret program.



Select: Analysis → Interpret



Select: Graph → Support Capacity Plots

Use the spin control to increase the number of envelopes to 3. The dialog should look like:

Support Capacity Plot  

Support Element:
 Tunnel Liner ▼

Factor of Safety Envelopes
 Number of Envelopes: 3 

#	Factor of Safety
1	1
2	1.2
3	1.4

Stages to Plot

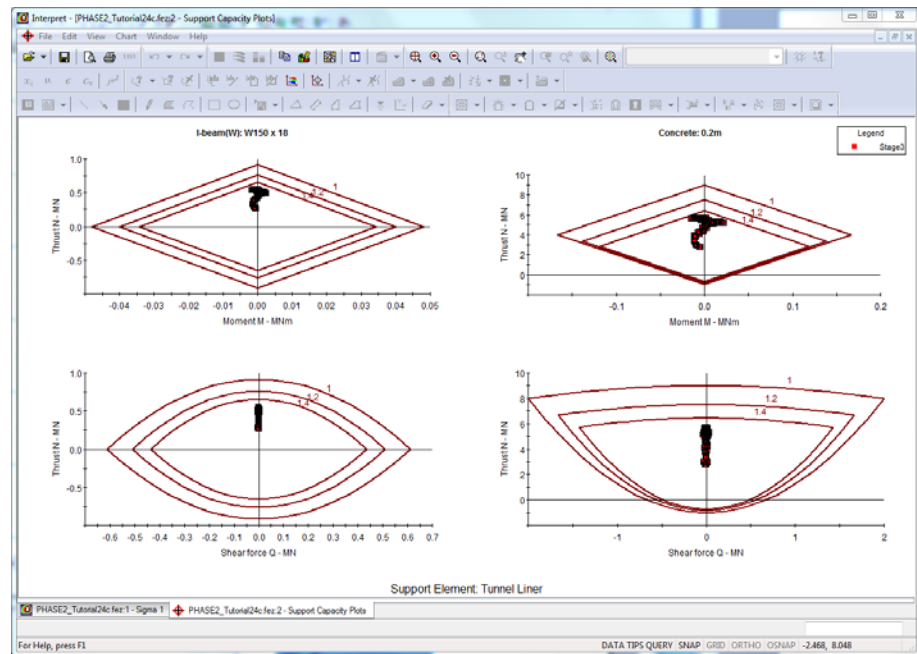
- ☐ Initial Stage
- ☐ Tunnel Relaxation
- ☒ Support Installed

Select All Clear All

OK Cancel

Press OK.

The following plot is generated.



Right away you'll notice that all the data points fall within the factor of safety=1.4 envelope, on all four plots. This means that the support system chosen has a factor of safety greater than 1.4 thus achieving the design factor of safety.

You can also use some of the tools previously demonstrated to filter out points with a factor of safety less than 1.4. You'll see that none exist.

This concludes the tutorial; you may now exit the *Phase2* Interpret and *Phase2* Model programs.